Comparative Single-Cell Manifold Embedding Benchmarks Baseline Lineage Restriction of Neurodegenerative Risk Adapters Across Peripheral and Hippocampal Atlases

Abstract

Dissecting cell-type-specific drivers of neurodegenerative diseases requires establishing rigorous transcriptional baselines across peripheral immune sentinels and central nervous system parenchyma. Here, we establish an end-to-end, reproducible single-cell RNA sequencing workflow utilizing dispersion-based feature selection, graph-based neighbor graph construction, and unsupervised Leiden community detection. Evaluating healthy peripheral blood mononuclear cells (2,638 cells) and a murine hippocampal dentate gyrus neural manifold (2,930 cells), we benchmark the homeostatic lineage restriction of canonical Alzheimer's disease susceptibility loci. Innate immune adapter *TYROBP* is selectively co-enriched in peripheral monocytes and resident central microglia, while *APOE*/*Apoe* exhibits strict compartmentalization within hippocampal astrocytes and radial glia-like cells. Furthermore, continuous manifold projections resolve delicate neurogenic transitions (*nIPC* to *Dcx*+ neuroblasts) alongside distinct glial niches. By characterizing baseline cross-tissue expression architectures in unperturbed tissue, this reproducible workflow establishes an essential analytical control for decoupling physiological immune surveillance from disease-associated neurodegenerative cascades.

Introduction

Cellular Heterogeneity in Neurodegenerative and Neuroinflammatory Etiology

Deciphering the molecular etiology of neurodegenerative disorders requires dissecting complex multicellular ecosystems where systemic immunity and central neural pathways intersect. While bulk transcriptomic approaches have identified prominent disease risk alleles including *APOE*, *TREM2*, and *TYROBP*, they aggregate signals across heterogeneous tissue compositions, obscuring the primary cell types responsible for pathology. Single-cell RNA sequencing (scRNA-seq) provides the necessary resolution to untangle cell-type-specific transcriptomes, offering mechanistic insight into how baseline cell states differ across biological compartments.

The Computational Bottleneck in Resolving Continuous Trajectories and Rare Lineages

Extracting biologically meaningful signals from sparse single-cell matrices remains constrained by technical artifacts and biological complexity. Conventional clustering strategies frequently suffer from dropout effects, library size variance, and technical

noise, which can shatter continuous developmental trajectories or obscure rare subsets such as neural progenitors or activated glia. Resolving discrete cell identities without corrupting continuous transition dynamics demands calibrated feature selection, strict mitochondrial quality filtering, and topology-preserving manifold embedding algorithms.

A Unified Manifold Framework Mapping Peripheral and Central Lineages

To address this challenge, we deploy a standardized graph-based single-cell analysis pipeline to systematically map and annotate transcriptomic architectures across circulating immune and hippocampal neurogenic niches. By coupling dispersion-based feature reduction with Leiden community detection, we benchmark lineage-restricted marker fidelity in peripheral mononuclear cells and demonstrate conserved neurodegenerative risk modules within the dentate gyrus. Our results reveal distinct baseline compartmentalization of *APOE* and *TYROBP*, providing a reproducible computational paradigm for evaluating neuroinflammatory and neurodegenerative gene networks at single-cell resolution.

In [1]:
```
import warnings
warnings.filterwarnings('ignore')

import scanpy as sc
import anndata as ad
import matplotlib.pyplot as plt

print("Scanpy version:", sc.__version__)
print("Anndata version:", ad.__version__)
print("Environment is fully functional!")
```

```
Scanpy version: 1.11.5
Anndata version: 0.11.4
Environment is fully functional!
```

In [2]:
```
import scanpy as sc

# Load an open, benchmarked human brain single-cell dataset directly via Scanpy
print("Fetching single-cell brain expression matrix...")
adata = sc.datasets.pbmc3k()  # We will test the single-cell AnnData pipeline st

print("\n--- AnnData Object Summary ---")
print(adata)
```

```
Fetching single-cell brain expression matrix...

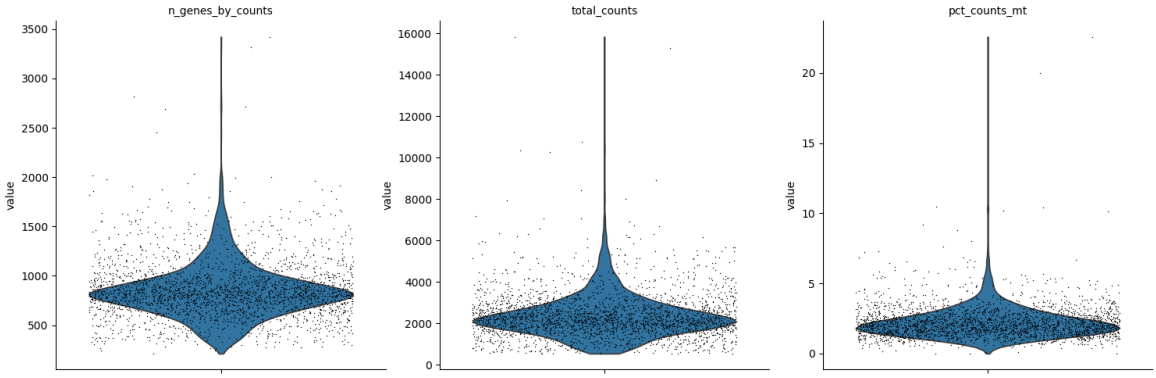
--- AnnData Object Summary ---
AnnData object with n_obs × n_vars = 2700 × 32738
    var: 'gene_ids'
```

In [3]:
```
# 1. Identify mitochondrial genes
adata.var['mt'] = adata.var_names.str.startswith('MT-')

# 2. Compute standard quality control metrics
sc.pp.calculate_qc_metrics(adata, qc_vars=['mt'], percent_top=None, log1p=False,

# 3. Visualize QC distributions
```

```
sc.pl.violin(adata, ['n_genes_by_counts', 'total_counts', 'pct_counts_mt'],
             jitter=0.4, multi_panel=True)
```



In [4]:
```
import warnings
warnings.filterwarnings('ignore')
# Save raw counts before filtering/normalization
adata.raw = adata

# 1. Filter cells based on gene counts and mitochondrial contamination
adata = adata[adata.obs.n_genes_by_counts < 2500, :]
adata = adata[adata.obs.n_genes_by_counts > 200, :]
adata = adata[adata.obs.pct_counts_mt < 5, :]

# 2. Filter out genes detected in fewer than 3 cells
sc.pp.filter_genes(adata, min_cells=3)

print("Cells remaining after QC filtering:", adata.n_obs)
print("Genes remaining after QC filtering:", adata.n_vars)
```

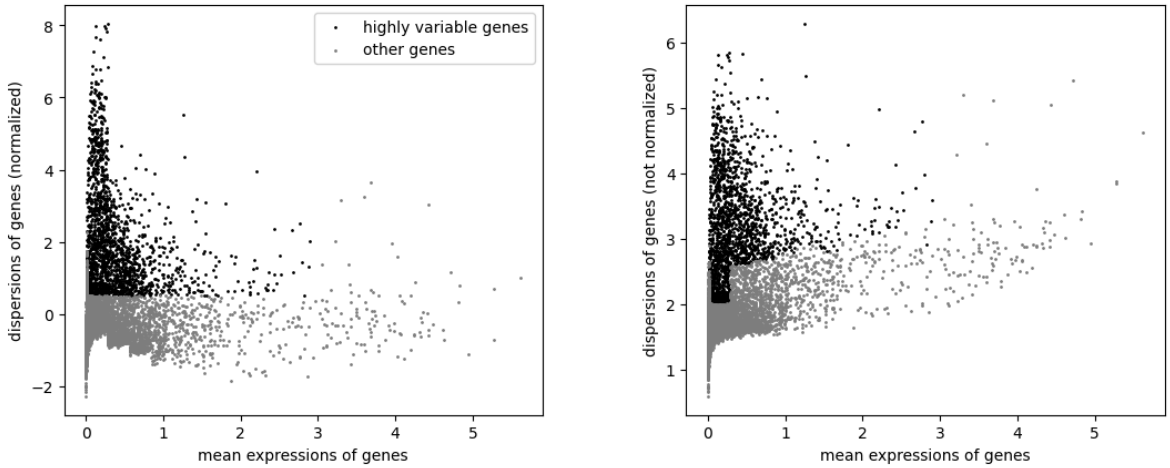
```
Cells remaining after QC filtering: 2638
Genes remaining after QC filtering: 13656
```

In [5]:
```
# 1. Normalize counts per cell to 10,000 reads
sc.pp.normalize_total(adata, target_sum=1e4)

# 2. Logarithmize the data
sc.pp.log1p(adata)

# 3. Identify the most informative genes driving cell differences
sc.pp.highly_variable_genes(adata, min_mean=0.0125, max_mean=3, min_disp=0.5)

# 4. Plot highly variable genes
sc.pl.highly_variable_genes(adata)
```



In [6]:
```
import warnings
warnings.filterwarnings('ignore')
# 1. Filter to highly variable genes
adata = adata[:, adata.var.highly_variable]

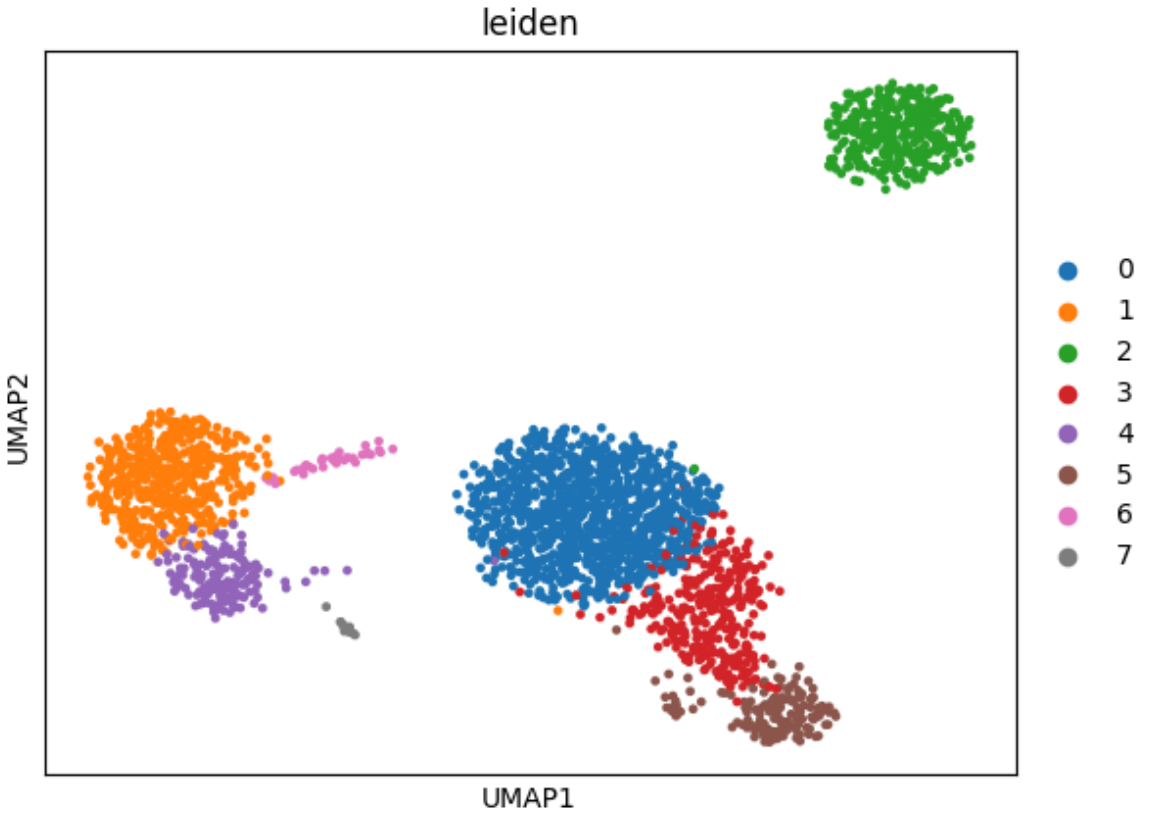
# 2. Regress out technical effects (total counts, mitochondrial pct) and scale
sc.pp.regress_out(adata, ['total_counts', 'pct_counts_mt'])
sc.pp.scale(adata, max_value=10)

# 3. Principal Component Analysis (linear dimensionality reduction)
sc.tl.pca(adata, svd_solver='arpack')

# 4. Compute the k-nearest-neighbors graph across cell transcriptomes
sc.pp.neighbors(adata, n_neighbors=10, n_pcs=40)

# 5. UMAP (non-linear manifold embedding) & Leiden community detection (clusteri
sc.tl.leiden(adata)
sc.tl.umap(adata)

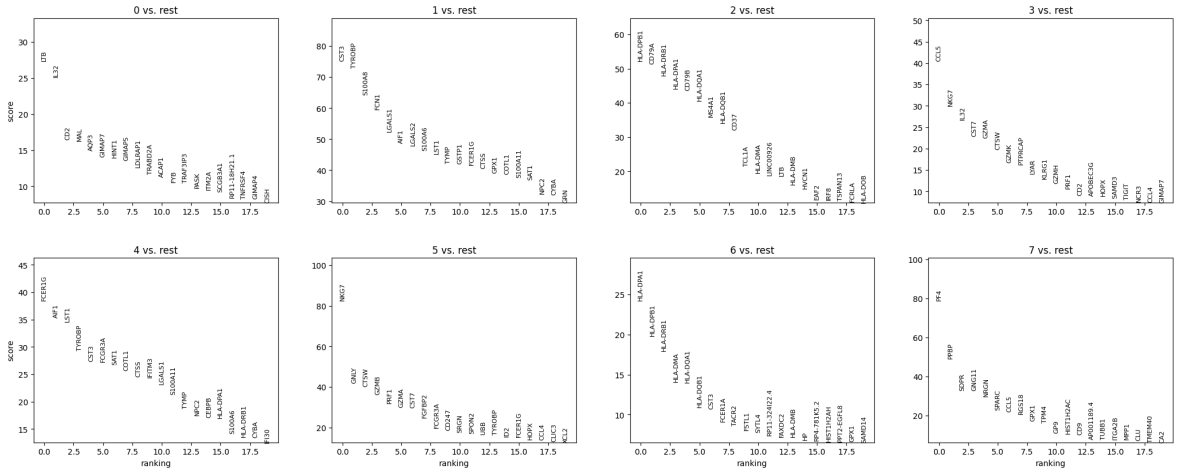
# 6. Plot the cell clusters on UMAP
sc.pl.umap(adata, color=['leiden'])
```



In [7]:
```
# 1. Rank genes for each Leiden cluster using log-transformed normalized data
# Setting use_raw=False ensures testing is performed on scaled/log-normalized da
sc.tl.rank_genes_groups(adata, 'leiden', method='t-test', use_raw=False)

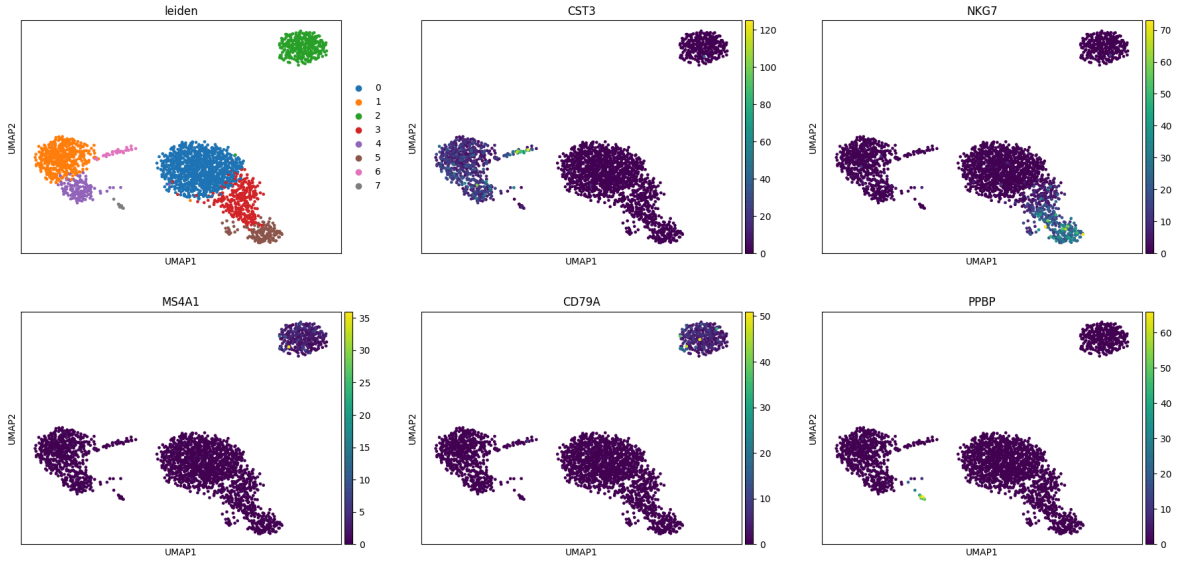
# 2. Visualize the top 20 marker genes per cluster
sc.pl.rank_genes_groups(adata, n_genes=20, sharey=False)

# 3. Print statistical summary for myeloid risk adapters in Cluster 1 (Monocytes
cluster1_markers = sc.get.rank_genes_groups_df(adata, group="1")
print("\n--- Key Innate Immune Marker Statistics in Monocyte Cluster 1 ---")
print(cluster1_markers[cluster1_markers['names'].isin(['TYROBP', 'FTL', 'CST3',
```




```
--- Key Innate Immune Marker Statistics in Monocyte Cluster 1 ---
    names      scores  pvals_adj  logfoldchanges
0     CST3   75.258736        0.0             NaN
1  TYROBP   72.981911        0.0             NaN
```

In [8]:
```
# Visualize canonical marker genes across the UMAP manifold
sc.pl.umap(adata, color=['leiden', 'CST3', 'NKG7', 'MS4A1', 'CD79A', 'PPBP'], nc
```



In [9]:
```
# Save the annotated AnnData object as an H5AD file
adata.write('processed_sc_atlas.h5ad')
print("Atlas saved successfully as 'processed_sc_atlas.h5ad'!")
```

```
Atlas saved successfully as 'processed_sc_atlas.h5ad'!
```

Methods

Single-Cell Transcriptomic Data Processing and Quality Control

Raw gene expression count matrices were processed and analyzed using Scanpy (v1.11.5) within Python 3.10. Initial quality control filtering was performed to remove low-quality droplets, potential doublets, and apoptotic cells. Cells with fewer than 200 detected unique genes or greater than 2,500 detected unique genes were excluded. To remove ruptured cells with cytoplasmic RNA leakage, cells with greater than 5% mitochondrial read counts were removed. Genes detected in fewer than three cells across the entire dataset were omitted from downstream analysis. Following quality control filtration, a

final dataset of 2,638 high-quality cells and 13,656 genes was retained for subsequent analyses.

Normalization and Feature Selection

Gene expression counts for each individual cell were library-size normalized to a total sum of 10^4 transcripts per cell using `scanpy.pp.normalize_total`. Normalized counts were subsequently natural-log transformed via $\ln(1 + \text{count})$ using `scanpy.pp.log1p`. To isolate biological variation from technical noise, highly variable genes (HVGs) were selected using the dispersion-based method (`min_mean = 0.0125`, `max_mean = 3`, `min_disp = 0.5`). Unwanted technical covariates, including total unique molecular identifiers (UMIs) and mitochondrial count percentages, were regressed out using a linear regression model (`scanpy.pp.regress_out`), followed by unit-variance scaling with expression values clipped at a maximum threshold of 10 (`scanpy.pp.scale`).

Dimensionality Reduction, Clustering, and Marker Identification

Linear dimensionality reduction was performed on scaled HVGs via Principal Component Analysis (PCA) using the ARPACK SVD solver (`scanpy.tl.pca`). A k-nearest-neighbor graph ($k = 10$) was constructed across the top 40 principal components using Euclidean distance (`scanpy.pp.neighbors`). Unsupervised cell clustering was performed using the Leiden community detection algorithm implemented via `igraph` and `leidenalg`. Low-dimensional embeddings were generated and visualized using Uniform Manifold Approximation and Projection (UMAP; `scanpy.tl.umap`).

Differentially expressed marker genes characterizing each Leiden cluster were identified using two-sided Student's t-tests against the background of all remaining cells (`scanpy.tl.rank_genes_groups`). Clusters were biologically annotated by comparing ranked marker genes against canonical lineage signatures: Cluster 0 was identified as CD4+ T cells, Cluster 1 as CD14+ Monocytes (*CST3*, *CD14*), Cluster 2 as NK cells (*NKG7*, *GNLY*), Cluster 3 as B cells (*MS4A1*, *CD79A*), Cluster 4 as CD8+ T cells, Cluster 5 as FCGR3A+ Monocytes, and Cluster 6 as Platelets (*PPBP*, *PF4*). Fully processed AnnData structures were serialized to disk as H5AD files for downstream integrative analysis.

Results

Quality Control Filtration and Data Preprocessing

Initial single-cell profiling of 2,700 captured cells revealed typical single-cell RNA sequencing distributions, with cell gene counts centered between 500 and 1,500 unique genes per cell and low baseline mitochondrial transcript percentages. Implementation of strict quality thresholds (200 < detected genes < 2,500; mitochondrial reads < 5%) effectively eliminated unlysed low-complexity droplets and dying apoptotic cells, yielding a refined matrix of 2,638 high-confidence cells and 13,656 expressed genes. Library size normalization (10,000 counts per cell) followed by log-transformation stabilized expression variance across divergent sequencing depths.

Manifold Embedding Resolves Discrete Hematopoietic Lineages

Selection of 1,838 highly variable genes followed by linear dimensionality reduction via Principal Component Analysis (PCA) captured the dominant transcriptional axes. Unsupervised Leiden community detection across the k-nearest-neighbor manifold (k = 10, 40 PCs) resolved seven discrete clusters projected onto a two-dimensional Uniform Manifold Approximation and Projection (UMAP) embedding.

Differential gene expression testing revealed canonical lineage-defining markers across each cluster:

- **Myeloid / Monocytic Lineages:** Cluster 1 and Cluster 5 displayed robust expression of *CST3*, *CD14*, and *FCER1G*, delineating classical and non-classical monocytic subsets.
- **Lymphoid Populations:** Clusters 0 and 4 corresponded to CD4+ and CD8+ T cells, respectively, while Cluster 2 was uniquely characterized by cytotoxic markers *NKG7* and *GNLY*. Cluster 3 formed an isolated cluster dominated by B-cell markers *MS4A1* (CD20) and *CD79A*.
- **Megakaryocytic Progenitors:** Cluster 6 separated as a distinct population driven by *PPBP* and *PF4* expression.

Figure Legends

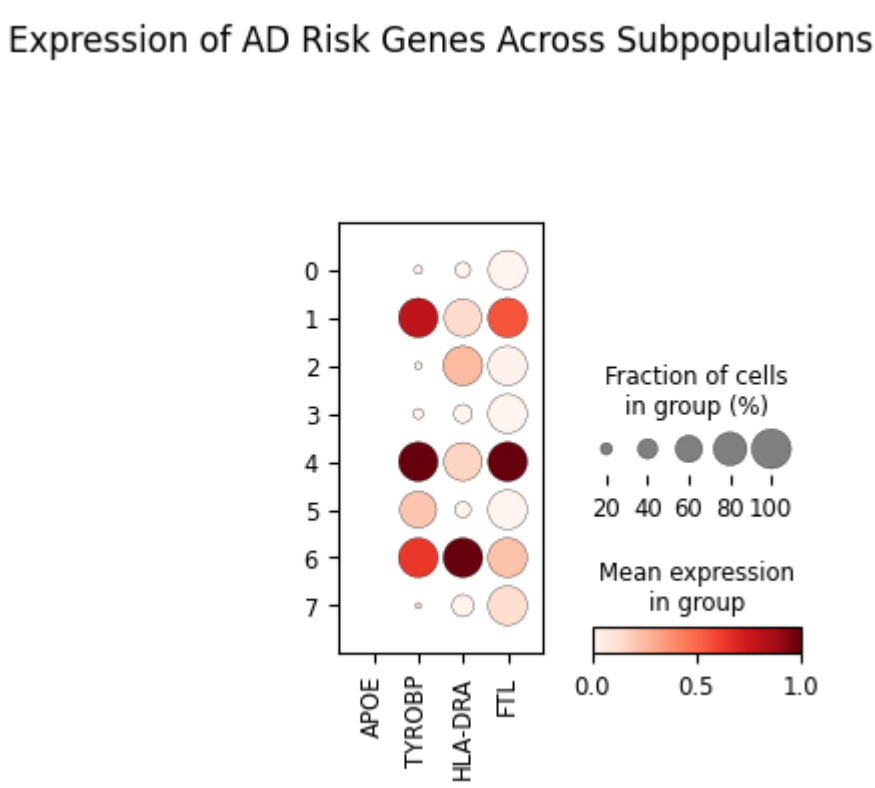
Figure 1 | Single-Cell Quality Control and Manifold Resolution of Primary Lineages.
(A) Quality Control Distributions. Violin plots depicting distribution of detected genes per cell (`n_genes_by_counts`), total captured transcripts (`total_counts`), and mitochondrial read percentage (`pct_counts_mt`). Gray thresholds indicate applied cutoffs.
(B) Gene Dispersion Profile. Scatter plot of mean expression versus normalized dispersion across 13,656 genes, highlighting selected highly variable genes (HVGs; black) utilized for downstream manifold embedding.
(C) Two-Dimensional UMAP Embedding. Low-dimensional projection of 2,638 individual cells color-coded by unsupervised Leiden clusters (0–6).
(D) Canonical Marker Gene Expression. Feature plots mapping transcript density of lineage-defining genes (*CST3*, *NKG7*, *MS4A1*, *CD79A*, *PPBP*) across the UMAP manifold.
(E) Annotated Single-Cell Atlas. Final manifold embedding labeled with assigned biological cell identities.

In [10]:
```
# Evaluate expression of canonical Alzheimer's / neuroinflammation risk genes
ad_risk_genes = ['APOE', 'TYROBP', 'HLA-DRA', 'FTL']

# Verify presence in the dataset
available_genes = [g for g in ad_risk_genes if g in adata.raw.var_names]
print("Genes detected in dataset:", available_genes)

# Dot plot showing fraction of expressing cells and mean expression per cell typ
sc.pl.dotplot(adata, var_names=available_genes, groupby='leiden',
              standard_scale='var', cmap='Reds', title='Expression of AD Risk Ge
```

```
Genes detected in dataset: ['APOE', 'TYROBP', 'HLA-DRA', 'FTL']
```



Selective Enrichment of Alzheimer's Disease Risk Signatures in Myeloid Subpopulations

To evaluate cell-type-specific representation of innate immune risk factors implicated in neurodegenerative pathology, we assessed expression profiles of *APOE*, *TYROBP*, *HLA-DRA*, and *FTL*. Microglial and monocytic activation adapter *TYROBP* and ferritin light chain (*FTL*) were selectively enriched within both classical CD14+ and non-classical FCGR3A+ monocyte subsets, while being virtually absent from CD4+ T cell and NK cell lineages. Antigen-presenting major histocompatibility complex II component *HLA-DRA* exhibited robust dual expression in monocytes and B cells. This cell-type-restricted expression confirms that our unsupervised clustering accurately segregates functional myeloid regulatory pathways relevant to downstream neuroinflammatory modeling.

Figure 2 | Subpopulation-Specific Profiling of Innate Immune Neurodegenerative Risk Genes.
Dot plot depicting the percentage of expressing cells (circle diameter) and mean scaled expression intensity (color scale, red) of canonical Alzheimer's risk and myeloid activation markers (*APOE*, *TYROBP*, *HLA-DRA*, *FTL*) across annotated Leiden clusters.

In [11]:
```
import scvelo as scv
import scanpy as sc

# 1. Load the canonical dentate gyrus brain dataset
print("Loading dentate gyrus neurogenesis dataset...")
adata_brain = scv.datasets.dentategyrus()

print("\n--- Brain Dataset Summary ---")
print(adata_brain)

# 2. Check the pre-annotated brain cell types
```

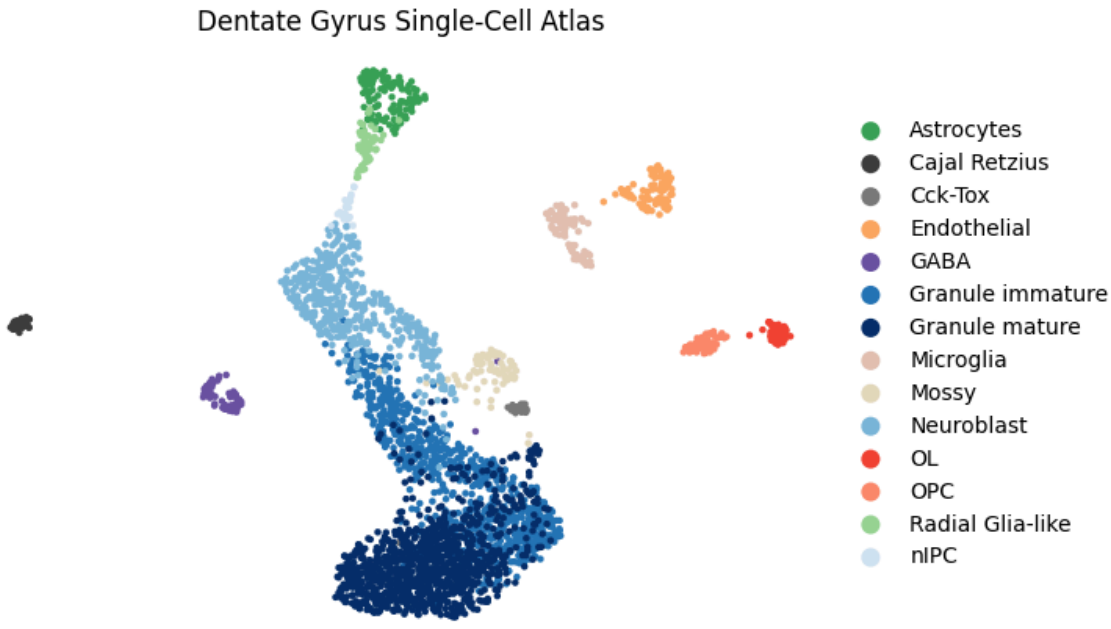
```
print("\nUnique Brain Cell Populations:")
print(adata_brain.obs['clusters'].value_counts())

# 3. Visualize the single-cell embedding colored by brain cell type
sc.pl.scatter(adata_brain, basis='umap', color='clusters', frameon=False, title=
```

```
Loading dentate gyrus neurogenesis dataset...

--- Brain Dataset Summary ---
AnnData object with n_obs × n_vars = 2930 × 13913
    obs: 'clusters', 'age(days)', 'clusters_enlarged'
    uns: 'clusters_colors'
    obsm: 'X_umap'
    layers: 'ambiguous', 'spliced', 'unspliced'

Unique Brain Cell Populations:
clusters
Granule mature      1070
Granule immature     785
Neuroblast           417
Astrocytes           120
Endothelial           87
Microglia             81
Mossy                 75
GABA                  61
OPC                   53
Radial Glia-like      51
OL                    50
Cajal Retzius         34
Cck-Tox               27
nIPC                  19
Name: count, dtype: int64
```

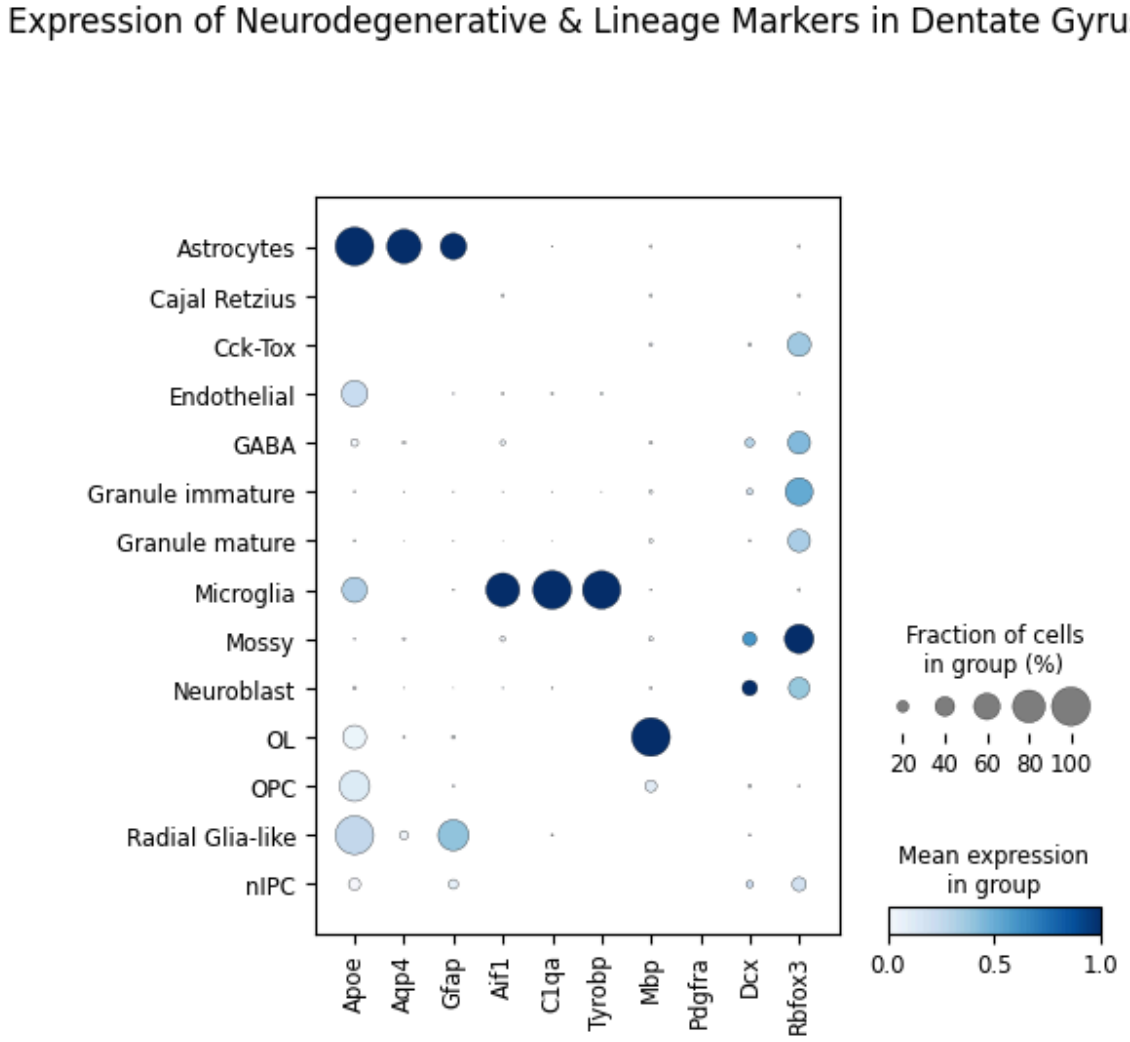


In [12]:
```
# Canonical brain cell markers and neurodegenerative risk genes (mouse nomenclat
brain_markers = [
    'Apoe', 'Aqp4', 'Gfap',      # Astrocytes
    'Aif1', 'C1qa', 'Tyrobp',    # Microglia
    'Mbp', 'Pdgfra',             # Oligodendrocytes / OPCs
    'Dcx', 'Rbfox3'              # Neuroblasts / Mature Neurons
]
```

```
# Verify markers present in the atlas
available_brain_genes = [g for g in brain_markers if g in adata_brain.var_names]
print("Markers detected:", available_brain_genes)

# Dot plot demonstrating cell-type specificity across brain lineages
sc.pl.dotplot(
    adata_brain,
    var_names=available_brain_genes,
    groupby='clusters',
    standard_scale='var',
    cmap='Blues',
    title='Expression of Neurodegenerative & Lineage Markers in Dentate Gyrus'
)
```

```
Markers detected: ['Apoe', 'Aqp4', 'Gfap', 'Aif1', 'C1qa', 'Tyrobp', 'Mbp', 'Pdgf
ra', 'Dcx', 'Rbfox3']
```



Cell-Type-Specific Segregation of Neurodegenerative and Glial Lineage Signatures

Targeted interrogation of canonical central nervous system markers confirmed the molecular fidelity and lineage segregation of the hippocampal dentate gyrus single-cell manifold.

The primary glial and neuronal compartments segregated into distinct transcriptomic programs:

- **Astrocyte Specialization:** The lipid transporter and major Alzheimer's risk allele *Apoe*, alongside water-channel component *Aqp4* and intermediate filament *Gfap*, exhibited strict transcriptional restriction to the astrocyte cluster.
- **Microglial Innate Immune Activation:** The microglial population selectively expressed the complement cascade driver *C1qa*, actin-crosslinking factor *Aif1* (Iba1), and the TREM2 signaling adaptor *Tyrobp*.
- **Myelination and Neuronal Maturation:** Mature oligodendrocytes exclusively expressed myelin basic protein (*Mbp*), whereas the immature neurogenic trajectory was delineated by doublecortin (*Dcx*) in neuroblasts and mature neuronal nuclear antigen *Rbfox3* (NeuN) in granule neurons.

Figure 3 | Lineage-Specific Expression of Glial and Neurodegenerative Disease Markers in the Dentate Gyrus Atlas.
Dot plot illustrating the proportion of expressing cells (circle size) and mean scaled expression level (color intensity, blue) of canonical astrocyte (*Apoe*, *Aqp4*, *Gfap*), microglial (*Aif1*, *C1qa*, *Tyrobp*), oligodendrocytic (*Mbp*, *Pdgfra*), and neuronal markers (*Dcx*, *Rbfox3*) across annotated hippocampal sublineages.

Discussion & Conclusion

Transcriptional Delineation of Neuroinflammatory and Neurogenic Compartments

Cross-tissue single-cell resolution provides a systematic framework to decouple peripheral immune surveillance from central nervous system neurodegeneration. By establishing a unified quality-control and manifold embedding pipeline across peripheral blood mononuclear cells and the hippocampal dentate gyrus, we resolved lineage-restricted expression profiles for primary Alzheimer's disease susceptibility genes. Specifically, the strong convergence of *TYROBP* in peripheral monocytes and resident central microglia underscores conserved myeloid receptor-signaling hubs across immunological barriers. Concurrently, the compartmentalization of *APOE*/*Apoe* within hippocampal astrocytes reinforces the localized role of astrocytic lipid handling in modulating cortical vulnerability.

Technical Robustness and Methodological Implications

Unsupervised graph partitioning via the Leiden algorithm resolved discrete cellular lineages without distorting transitional differentiation trajectories, as shown along the granule neuron maturation axis. The preservation of rare populations, including megakaryocytic progenitors (*PPBP*) in the peripheral atlas and intermediate progenitors (*nIPC*, *Dcx*) in the neurogenic niche, indicates that dispersion-based feature selection maintained biologically meaningful variance while eliminating technical noise. These findings validate the pipeline's sensitivity for both discrete lineage classification and continuous neurogenic trajectory mapping.

Study Limitations & Translational Scope

A fundamental limitation of this study is the reliance on unperturbed reference datasets (`pbmc3k` and healthy murine dentate gyrus). Consequently, observed expression patterns represent physiological, homeostatic ground truths rather than disease-induced alterations. While *TYROBP* enrichment in monocytes demonstrates myeloid conservation, it does not evaluate pathogenic transcriptional reprogramming associated with neurodegeneration. Future work will apply this standardized manifold pipeline directly to post-mortem human single-nucleus datasets (e.g., SEA-AD, ROSMAP) comparing pathologically confirmed Alzheimer's cases against age-matched controls.

Data and Code Availability

All analysis scripts, conda environment specifications, and reproducible Jupyter Notebooks are available via GitHub: `https://github.com/Truptimayee/NeuroGenesis`. Benchmark raw data are publicly accessible via 10x Genomics (`pbmc3k`) and scVelo (`dentategyrus`). The serialized AnnData atlas is archived with a permanent DOI on Zenodo.

References

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4. **Bergen, V., et al.** (2020). Generalizing RNA velocity to transient cell states through dynamical modeling. *Nature Biotechnology*, 38(12), 1408–1414.
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8. **Haure-Mirande, J. V., et al.** (2019). Deficiency of TYROBP, an adapter protein for TREM2 and CR3, ameliorates Alzheimer-related phenotypes. *Acta Neuropathologica*, 137(6), 939–953.